

Workflow of the I&R System

Table of content

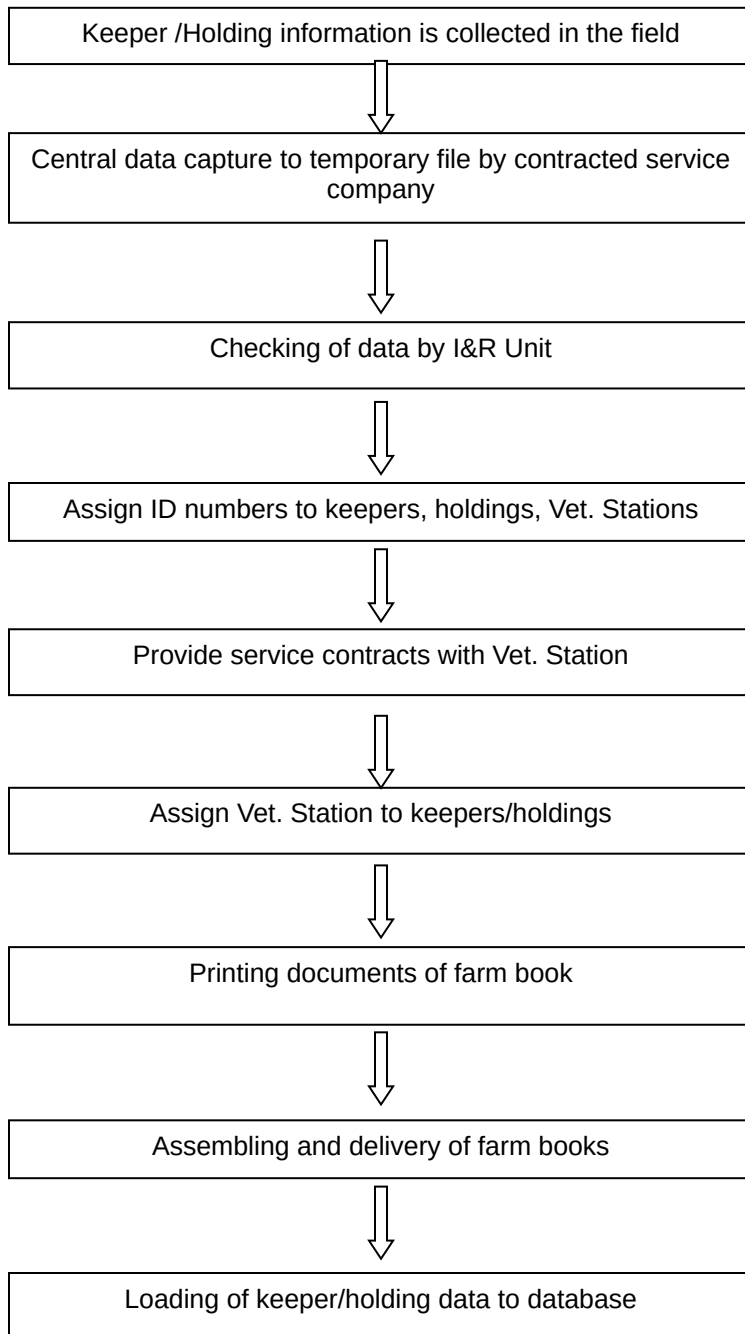
Glossary.....	3
Instance no. 1: Farm Census.....	4
Instance no. 2: Maintenance of Keeper/Holding addresses – New keeper/holding.....	6
Instance no. 3: Changing keeper/holding information.....	8
Instance no. 4: First Allocation of Ear Tags.....	10
Instance no. 5: Routine Allocation of Ear Tags.....	11
Instance no. 6: Ordering and distribution of replacement Ear Tags.....	12
Instance no. 7: Withdrawal of the ear tags.....	13
Instance no. 8: Notification of births.....	14
Instance no. 9: Routine registration and tagging on the spot.....	15
Instance no. 10: First tagging.....	16
Instance no. 11: Issuing of Cattle Passports.....	17
Instance no. 12: Communication of movements/death-on-farm on spot by PDA.....	18
Instance no. 13: Communication of movements/death-on-farm by Postcards.....	20
Instance no. 14: Communication of Slaughter at slaughterhouse.....	21
Instance no. 15: Movement communication at livestock markets and fairs.....	22
Instance no. 16: Communication of movements to alpine grazing areas.....	24
Instance no. 17: Re-prints of forms in case of consumption/loss.....	25
Instance no. 18: On spot inspections.....	26
Instance no. 19: Importation of animals from countries following EU rules.....	28
Instance no. 20: Importation of animals not intended for slaughter from 3 rd countries.....	30
Instance no. 21: Exportation of animals.....	32
Instance no. 22: Error correction procedure: spotted in field.....	33
Instance no. 23: Error correction procedure: spotted centrally at data entry (a priori plausibility checks).....	34
Instance no. 24: Error correction procedure: spotted centrally at regular intervals (posteriori plausibility checks).....	35
Instance no. 25: The I&R Archive.....	36

Glossary

BIP	Border Inspection Post
CPC	Central Processing Centre
VI	Veterinary Inspection/Veterinary Inspector
VS	I&R Service Unit (Veterinary Stations)

Instance no. 1: Farm Census

Working scheme of the execution of farm census



Description of farm census execution

The VD provides the census form to the VS. The census is carried out by staff of the VS. The data collection is based on the census form. The completed census form is sent by the VS to VD.

A contracted service company will key in the data of the census form. After all forms are keyed in, the data is checked and analysed by the VD in order to avoid double registering of keepers and holdings.

The VD assigns unique ID numbers to keepers and their holding(s) to all other system partner

- 1) Further holdings as village pastures and alpine grazing areas
- 2) Veterinary Stations
- 3) Veterinarians
- 4) Technicians
- 5) Livestock markets
- 6) BIPs
- 7) Veterinary Inspections
- 8) Veterinary Inspectors
- 9) Slaughter houses
- 10) Trader

according to the holding numbering system. After that the record of each keeper /holding is completed with the number of the responsible VS.

After that the printing of the documents of the farm book, the assembly of the farm book and the delivery to the VS is carried out.

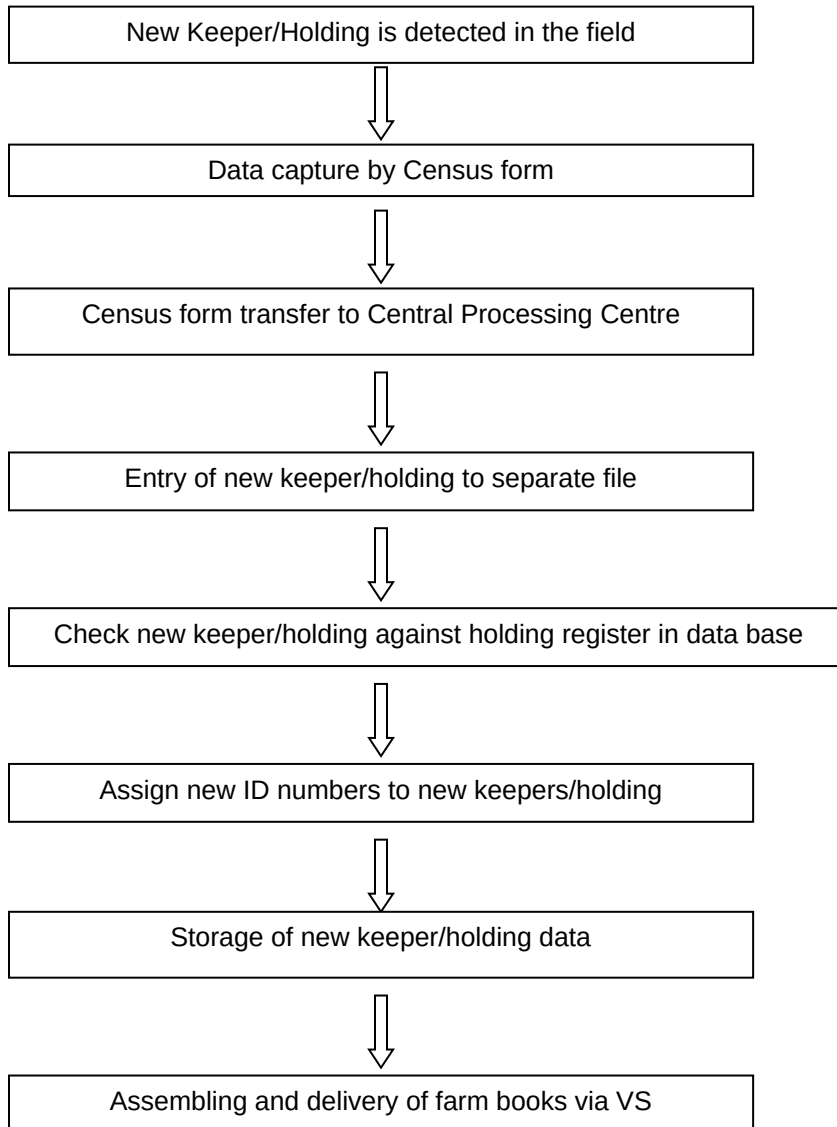
The records of the keeper/holdings and their VS are loaded to the central database.

Prerequisites

- 1) Contracting of VS for data collection in the field is done
- 2) Collection forms are designed, printed and distributed to the VS
- 3) Training of collection staff is done
- 4) Tendering, awarding and contracting of routine data entry is done
- 5) Tendering and awarding of printing, assembling and delivery of farm book is done
- 6) The central database is operational and ready to load the records

Instance no. 2: Maintenance of Keeper/Holding addresses – New keeper/holding

Working scheme for registering of a new holding



Description of registering of a new holding

New Holding is detected by VS field staff. Data capture is done by Census form. Data transfer of new holding to CPC is done.

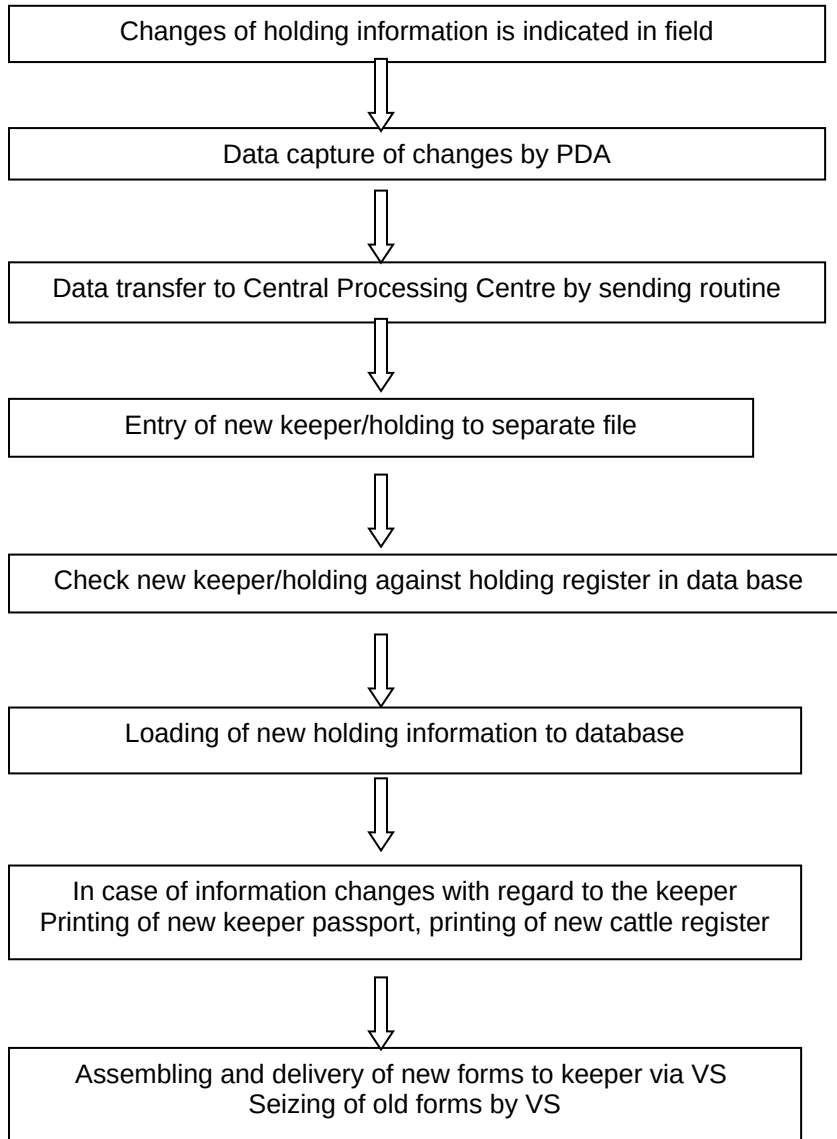
Keeper address is checked whether already existing, in case that no: new holding ID no. is allocated to the keeper and new holding is loaded to the data base. Forms for Farm book are printed and assembled. Farm book is dispatched via VS to the new keeper.

Prerequisites

- 1) The central database is operational and contains holdings for comparison
- 2) The central database provides application for comparison with existing holdings

Instance no. 3: Changing keeper/holding information

Working Scheme of changing keeper/holding information



Description of changing holding information

In case that the holding information is changed, the holding ID no. as a spatial unit remains unchanged. Changes of entered to the PDA and transferred to the CPC via routine data exchange. In Case of changes on keeper information, any form containing this information (Keeper Passport, Keeper tape sheet, cattle register for purchased animals) is printed and sent to the keeper via VS.

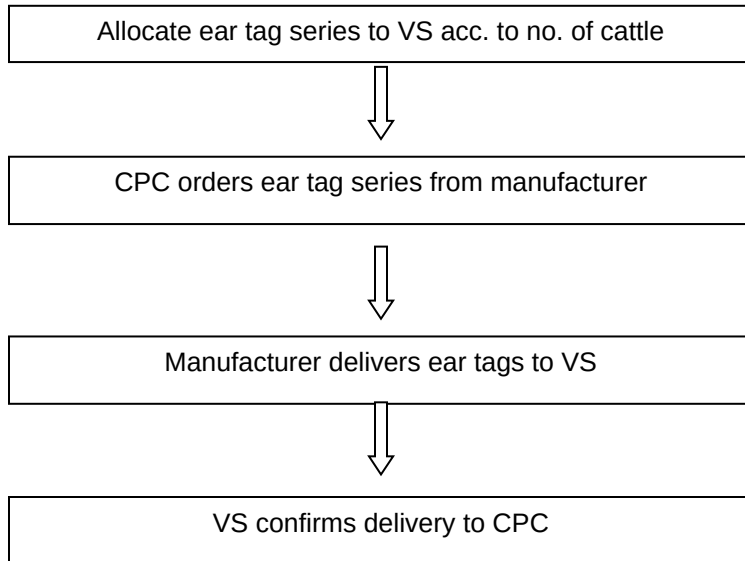
Prerequisites

- 1) The central database is operational and contains holdings for comparison

- 2) An application for performing changes and providing prints of required forms exist

Instance no. 4: First Allocation of Ear Tags

Working scheme of first allocation of ear tags



Description of first allocation of ear tags

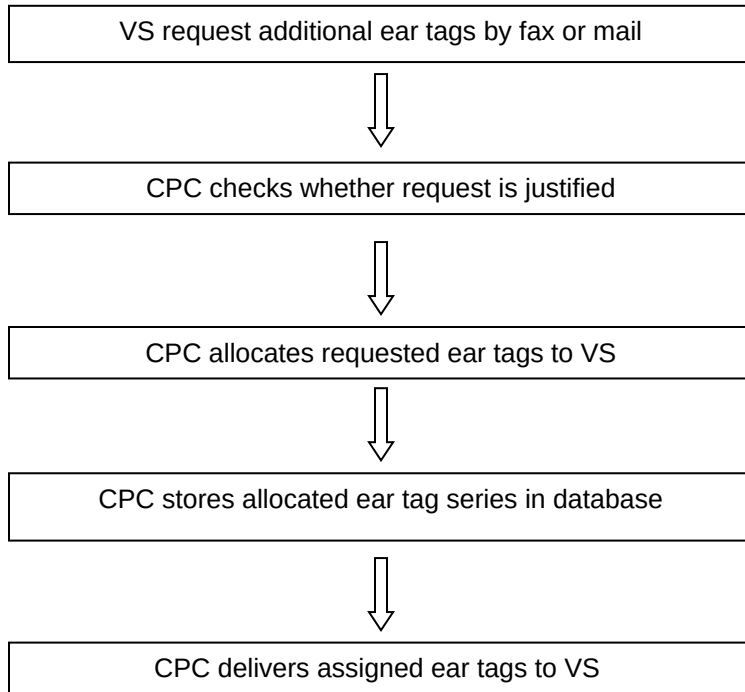
The CPC allocates ear tags to the contracted VS, according to the estimated demand for the first tagging. The allocated ear tags series per VS are ordered from manufacturer. The manufacturer directly ships ear tags to the VS. A confirmation of reception is forwarded to CPC

Prerequisites

- 1) Contracting of the I&R service in the field is done
- 2) Tendering and awarding of ear tags and tagging receipts manufacturing is done
- 3) Training of collection staff is done
- 4) Tendering and awarding of printing, assembling and delivery of farm book is done
- 5) The central database is operational and contains application for ear tag allocation

Instance no. 5: Routine Allocation of Ear Tags

Working scheme of routine allocation of ear tags



Description of routine allocation of ear tags

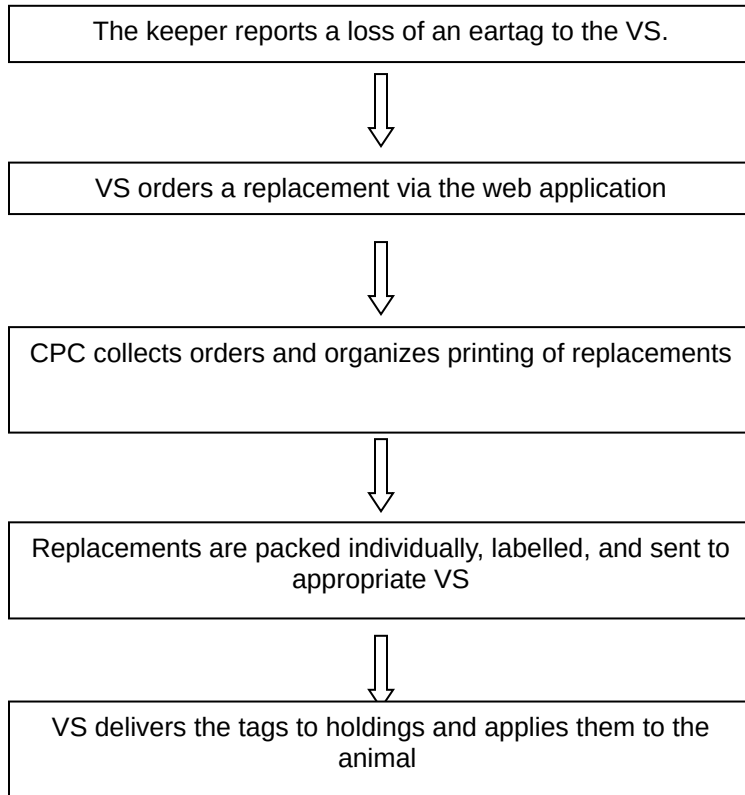
The VS requests additional ear tags from the CPC. The CPC allocate the ear tags series to the contracted VS, according to the estimated demand for ½ year. The allocated ear tags per VS are stored in the central database. The ear tags are shipped to the VS.

Prerequisites

- 1) Tendering and awarding of the I&R service in the field is done
- 2) Tendering and awarding of ear tag and form manufacturing is done
- 3) Training of collection staff is done
- 4) The central database is operational and contains application for ear tag allocation

Instance no. 6: Ordering and distribution of replacement Ear Tags

Working scheme of routine allocation of ear tags



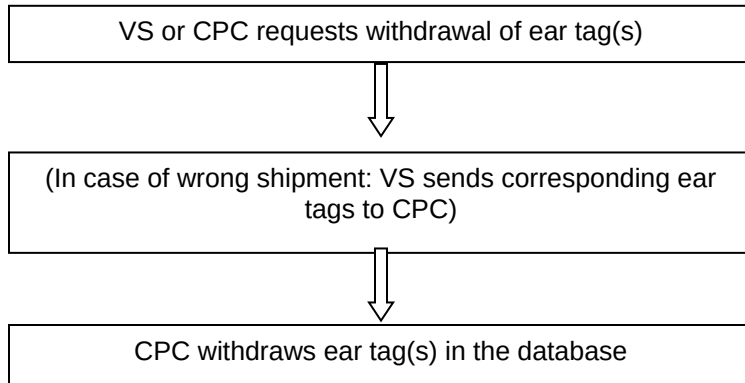
Description of routine allocation of ear tags

The keeper reports the loss of an eartag (animal number, single/pair). The VS collects requests and orders replacements via the web application. The CPC organizes printing of eartags weekly. The tags are individually packed and labelled by identification number, bar code, holding number and holding address. The individual packages are shipped to the VS. The VS delivers the replacements to the holding and performs the application to the animal ear.

- 5) Tendering and awarding of the I&R service in the field is done
- 6) Tendering and awarding of ear tag and form manufacturing is done
- 7) Training of collection staff is done
- 8) The central database is operational and contains application for replacement ear tag ordering

Instance no. 7: Withdrawal of the ear tags

Working scheme of routine withdrawal of ear tag(s)



Description of routine withdrawal of ear tags

In case that an ear tag (-ear tags) allocated VS cannot be used for tagging (damage of ear tag, loss of ear tag by VS, use for demonstration purposes, insufficient quality etc.), VS requests withdrawal of ear tag(s) from the system.

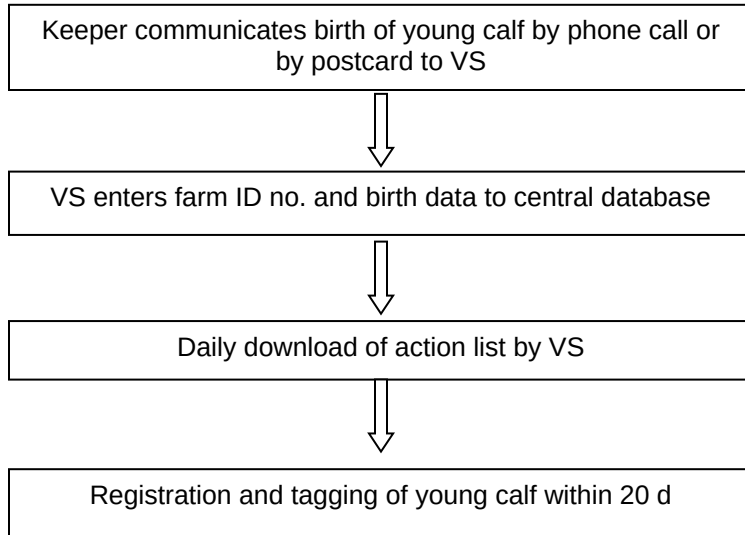
The CPC withdraws the relevant animal ID no(s). Thereafter, the animal ID no(s) is blocked. VS are not allowed to exchange ear tags directly between themselves.

Prerequisites

- 1) Tendering and awarding of the I&R service in the field is done
- 2) Tendering and awarding of ear tag and form manufacturing is done
- 3) Training of collection staff is done
- 4) The central database is operational and contains application for withdrawal of ear tags

Instance no. 8: Notification of births

Working scheme of birth notification



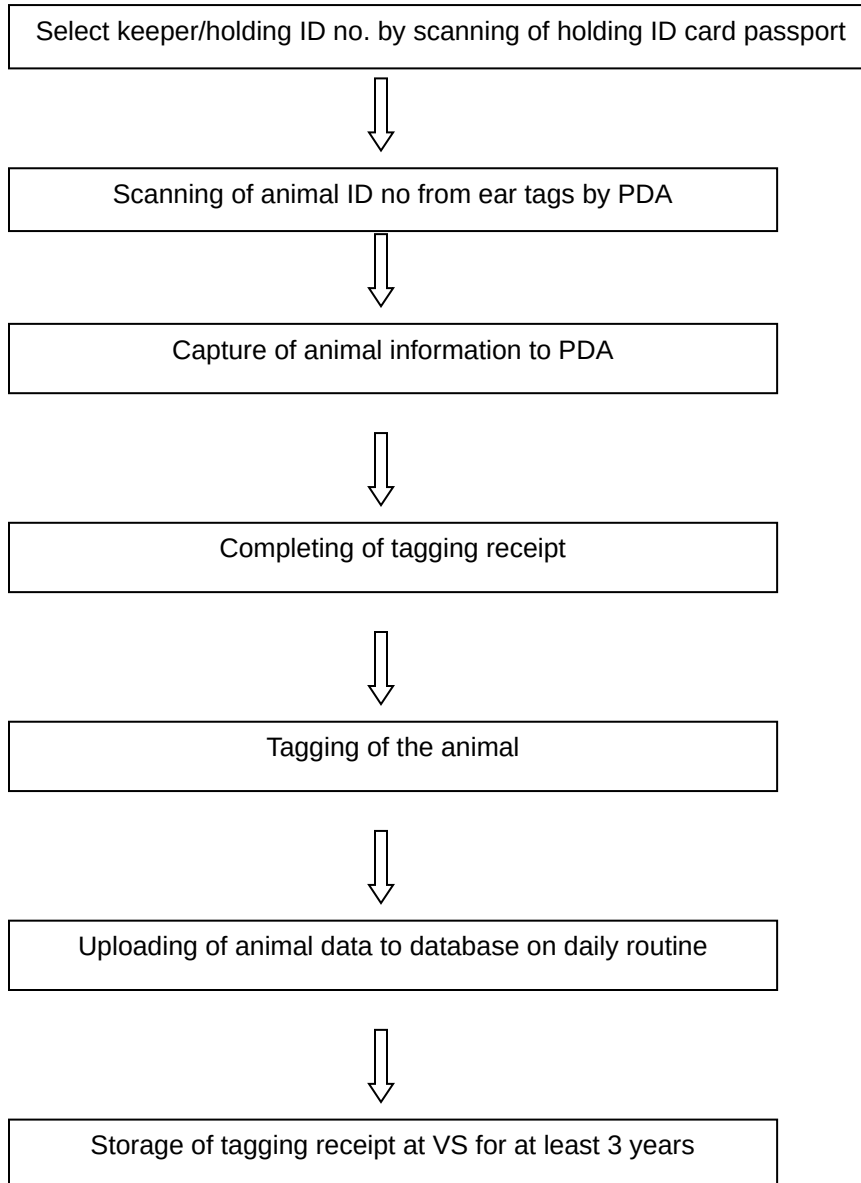
Description of birth notification

The farmer communicates birth event to VS within 7 days. The event is entered to database by VS staff. Daily download of action lists indicate the farms to be visited for registration and tagging. Registration is to be done within 20 days after birth. In case of missing registration after 20 days an alert is provided by the database.

The software provides a daily list of nonregistered calves sorted by villages and VS; the software provides a list of calves not registered in due time (20 days of age).

Instance no. 9: Routine registration and tagging on the spot

Working scheme of Routine registration and tagging on the spot

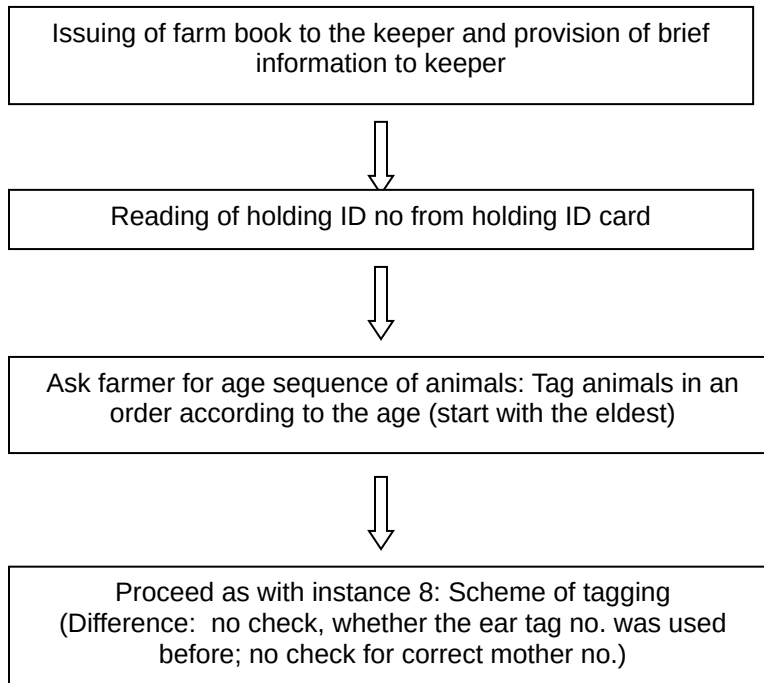


Description of routine tagging on the spot

The veterinarian captures the holding information to PDA by reading the barcode of the holding ID card. The animal information is entered to PDA. The animal ID no on both ear tags is scanned by the PDA-scanner and checked. After that, the veterinarian completes the tagging receipt and tags the animal. The data is uploaded to central data base on daily routine work. The tagging receipts are stored at VS for at least 3 years.

Instance no. 10: First tagging

Working scheme of first tagging

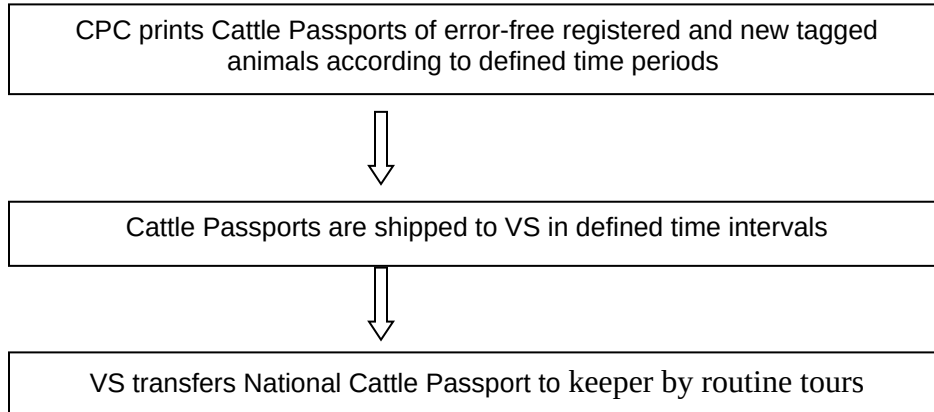


Description of first tagging

First tagging is executed on the spot and always is accompanied by full registration of the animal to the PDA database. The ear tags are scanned by the PDA scanner. The veterinarian captures the animal information by the PDA. In case of tagged animals moved to the farm prior to the farm visit, an on-farm movement of the animals is performed. In case of animals from the pilot project, the animals keep their ear tags and the animal ID no. is entered by hand to the PDA. At VS the PDA is connected to the central database and the animal information is uploaded to the central database.

Instance no. 11: Issuing of Cattle Passports

Working scheme of issuing of Cattle Passports

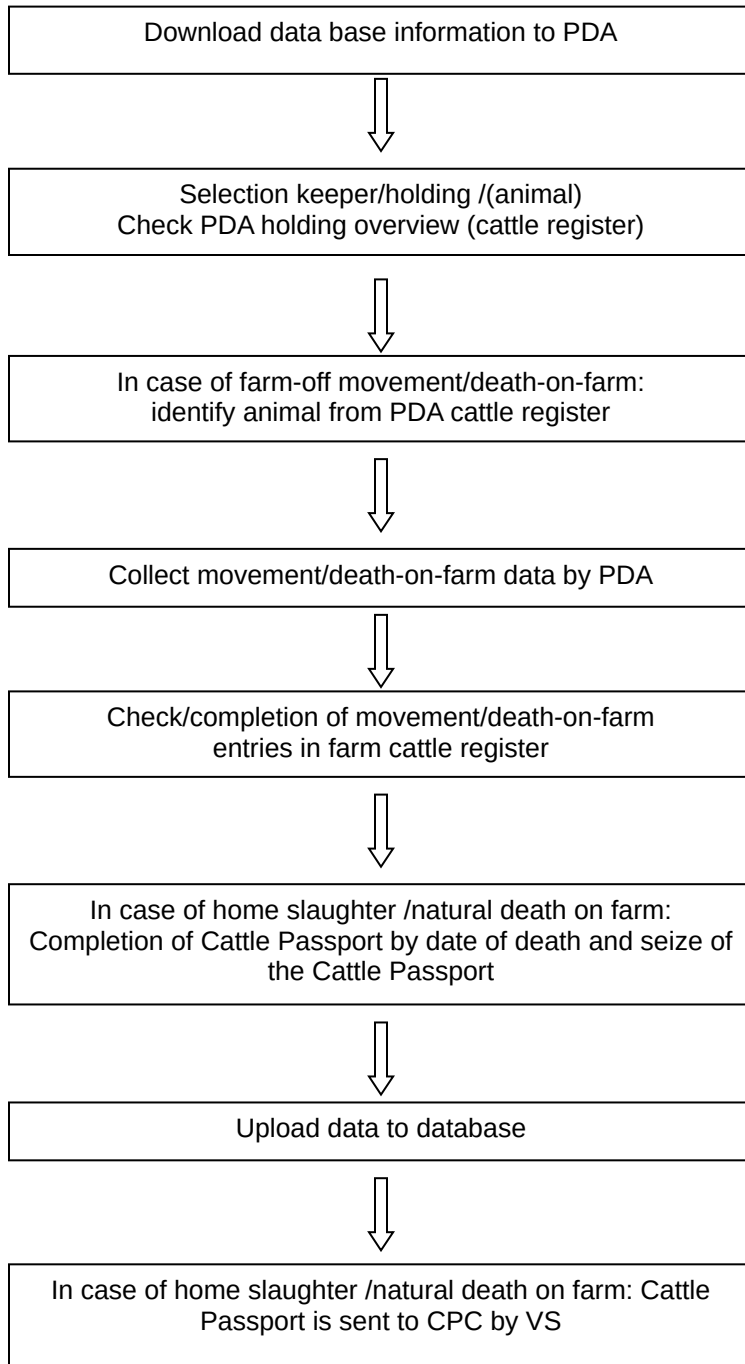


Description of issuing of Cattle Passports

CPC prints Cattle Passports of error-free registered and new tagged animals according to defined periods. Cattle Passports are parcelled and shipped to VS. VS forwards documents to keeper according to routine circulation

Instance no. 12: Communication of movements/death-on-farm on spot by PDA

Working scheme of communication of movements/death-on-farm on spot by PDA



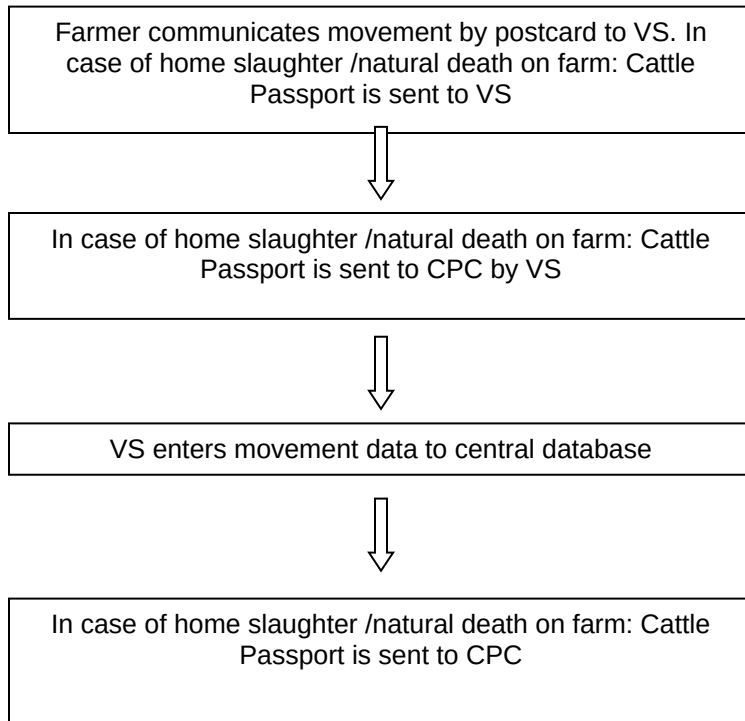
Description of communication of movements/death-on-farm by PDA

The PDA is loaded with the records of the keepers/holding and animals. The veterinarian compares on each farm visit the cattle register stored in the PDA with the animals on the farm. If a movement (on / off) has occurred, that is not yet stored in the PDA, the collection of movement data is carried out by the veterinarian. The animal number, previous/next holding and date and reason of movement are entered to the PDA. The data is uploaded to data base.

In case of death on farm (e.g. home slaughter) the vet completes the Cattle Passport by date of death and sends the Cattle passport to the CPC. CPC stores the cattle passport for at least 3 years.

Instance no. 13: Communication of movements/death-on-farm by Postcards

Working scheme of communication of movements/death-on-farm by postcards

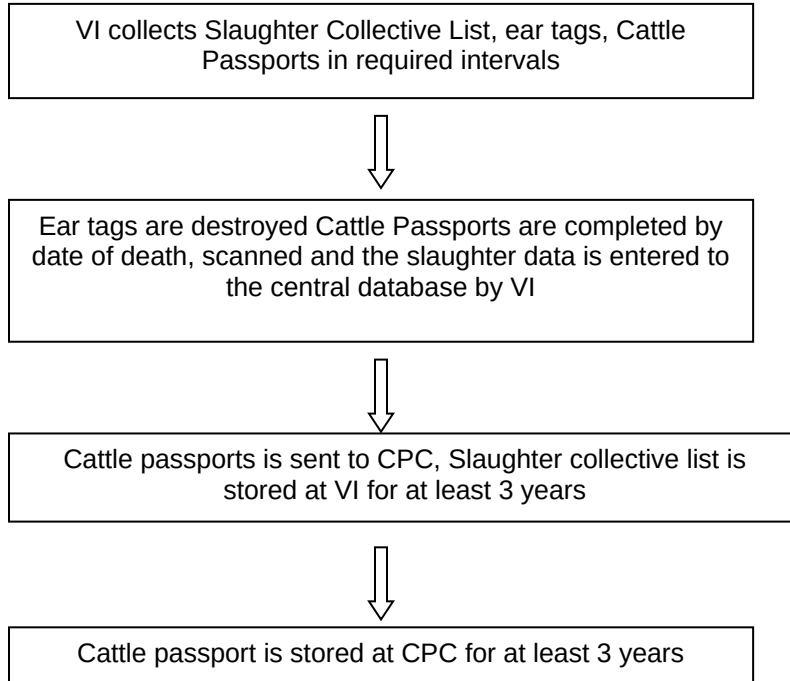


Description of communication of movements/death-on-farm by postcards

Farmer communicates movement by postcard to VS. In case of home slaughter /natural death on farm Cattle Passport is sent to VS as well. Data capture is done by direct application on central database. In case of death on farm (e.g. home slaughter) VS completes the Cattle Passport by date of death and sends the Cattle passport to the CPC. CPC stores the cattle passport for at least 3 years.

Instance no. 14: Communication of Slaughter at slaughterhouse

Working scheme of communication of Slaughter at slaughterhouse

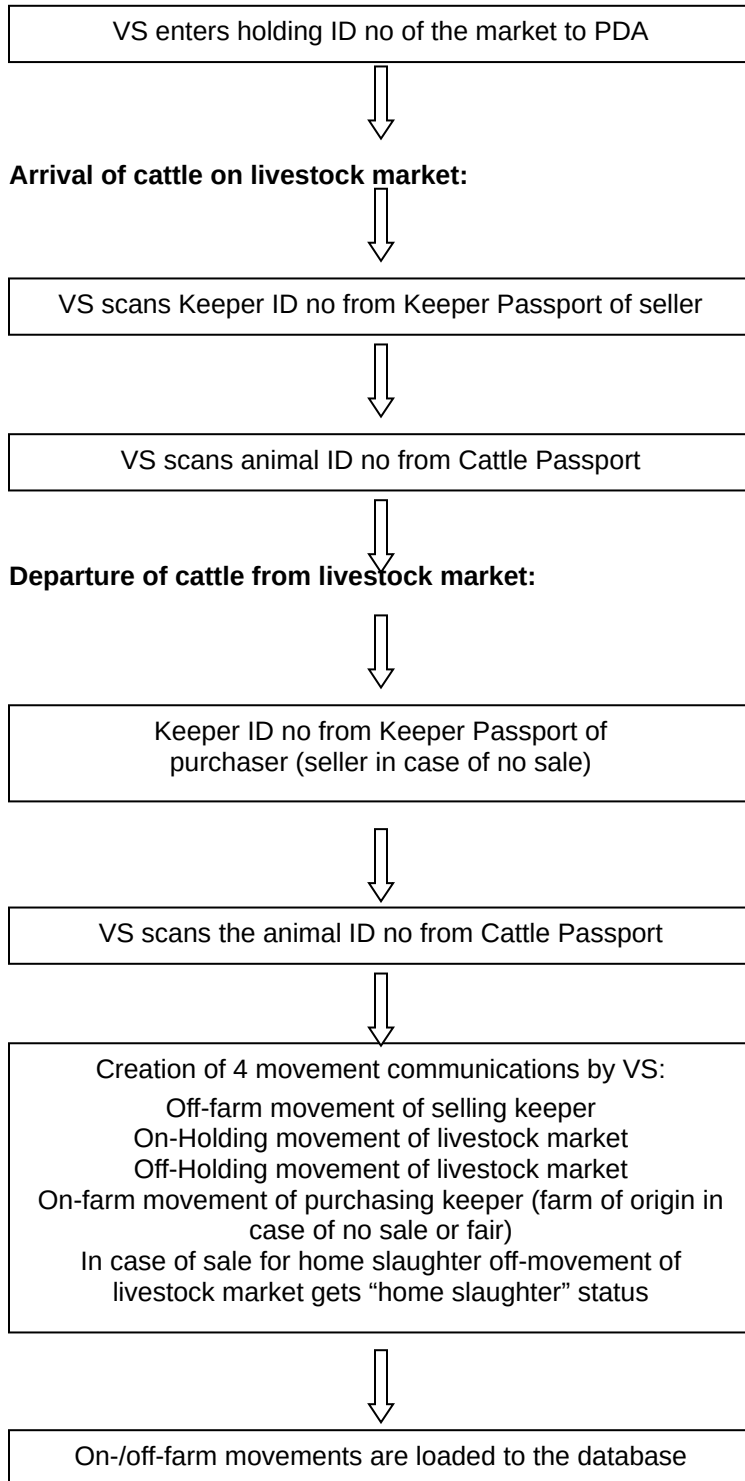


Description of recording slaughtering at slaughter-house

The veterinary inspector (VI) at the slaughterhouse collects Slaughter Collective List, ear tags and corresponding Cattle Passports. The ear tags are destroyed, Cattle Passports are completed by date of death by VI. The animal's slaughter data is fed to the central data base at VI. Cattle Passports are sent to CPC. The cattle passport is stored at CPC for at least 3 years.

Instance no. 15: Movement communication at livestock markets and fairs

Working scheme of movement communication at livestock markets and fairs



Description of movement communication at livestock markets and fairs

The Vet provides services for communication of movements on livestock markets and fairs within his region of competence. When the seller enters to the market place, the Vet scans the Keeper ID no from the Keeper passport of the seller and the animal ID no(s) from Cattle Passports of all his animals by the PDA-scanner. The same procedure is applied with the purchaser when the animals leave the market place.

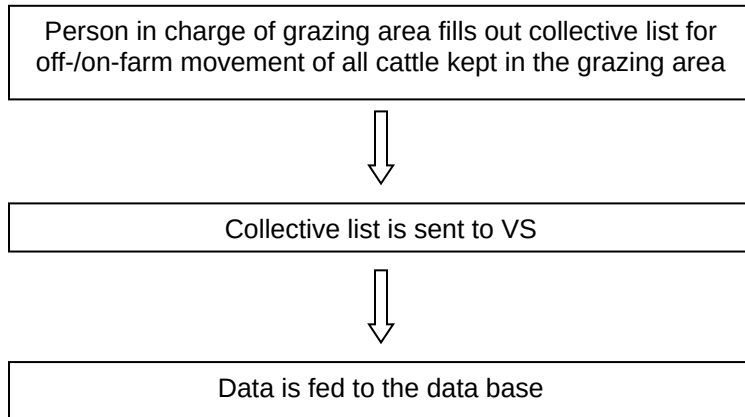
The PDA-software creates automatically an off-holding movement from the selling farm and an on-holding movement to the livestock market as well as an off-holding movement from the livestock market and an on-holding movement to the purchaser. In case that the keeper is not identical with the holding, the holding ID no as shown on the Cattle Passport must be used.

With regards to cattle origin from all Macedonian regions no further checks are carried out.

In case of sale for private slaughter off-movement of livestock market gets "home slaughter" status. If the animal is not sold or in case of fairs, the Vet Scans the ID no of the keeper of origin again and creates an on-farm movement to the farm of origin.

Instance no. 16: Communication of movements to alpine grazing areas

Working scheme of communication of movements to alpine grazing areas

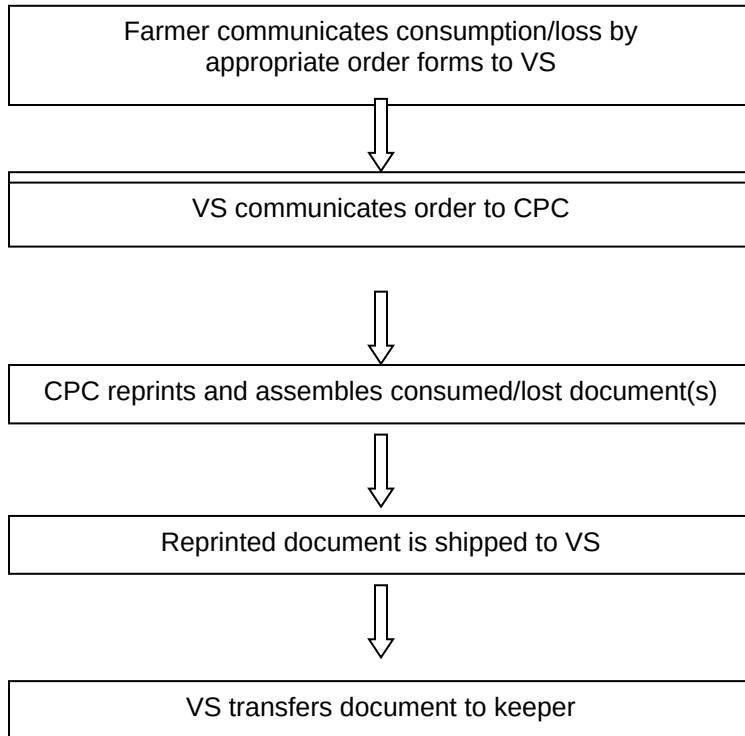


Description of communication of movements to alpine grazing areas

Movements are reported by person/organisation in charge of the grazing area (e.g. Central Pasture Agency) by collective lists containing all cattle in a separated area. The collective list is sent to VS, where the data is fed to the data base

Instance no. 17: Re-prints of forms in case of consumption/loss

Working scheme of re-prints of forms in case of consumption/loss



Description of re-prints of forms in case of consumption/loss

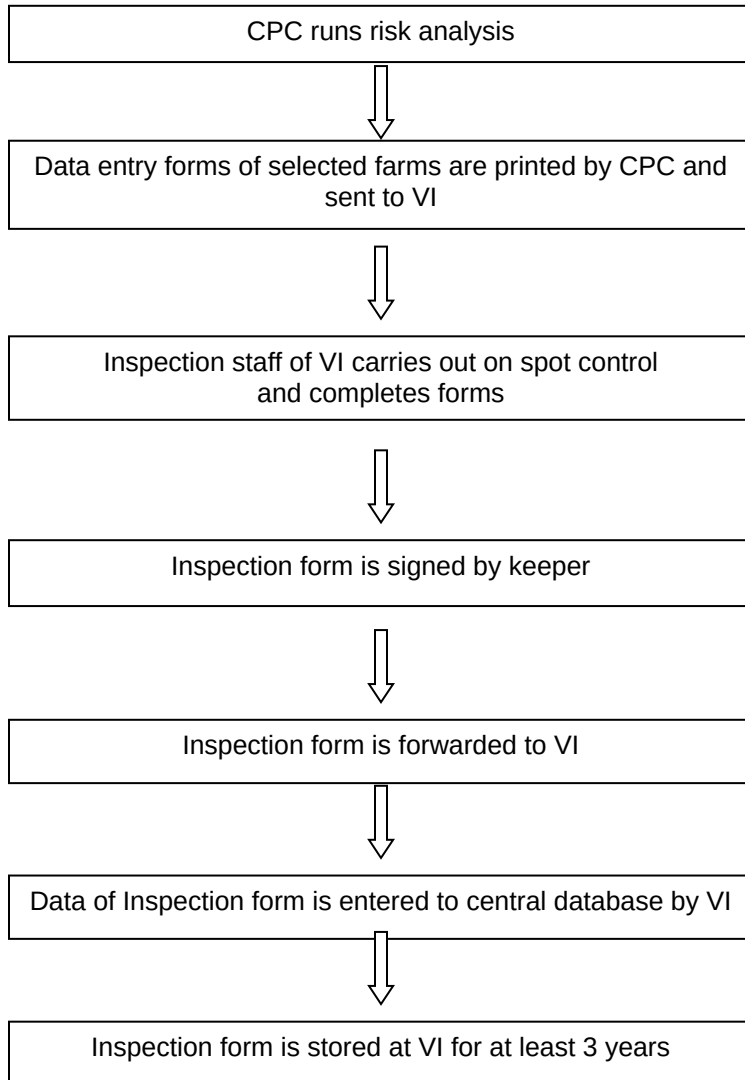
The keeper informs VS about consumed/lost document by appropriate order form. This form refers to:

- Order form
- Farm book
- Cattle Passport
- Cattle Register
- Holding ID cards
- Movement Cards
- Birth notification
- Ear tag

The order is communicated to CPC by VS. The Document is reprinted and – in case of ear tags and passports an additional roman no indicates the no of repeats. In case of farm book – this is assembled by CPC and the document is shipped to farmer through VS.

Instance no. 18: On spot inspections

Working scheme of on-spot inspections



Description of on-spot inspections

CPC runs a risk analysis in order to determine which of the 10% of keeper/holding are to be inspected. The information of selected keepers/holdings is on data entry forms for farm inspection. The data entry form is sent to the responsible VI. The VI carries out on spot control and completes the entry form with the results of inspection. The keeper has to sign the data entry form. The entry form is entered to the central database at VI and stored at VI for at least 3 years.

Software support

An on-the-spot control software support is provided. It consists of the following elements:

1. Risk analysis

[run by hand]

2. Random pick of farms

An application is provided to pick a specified number of holdings within a specified region (randomly according to _____ methodology)

3. Random pick of animals

An application is provided to pick a specified number of animals within a specified region (randomly according to _____ methodology)

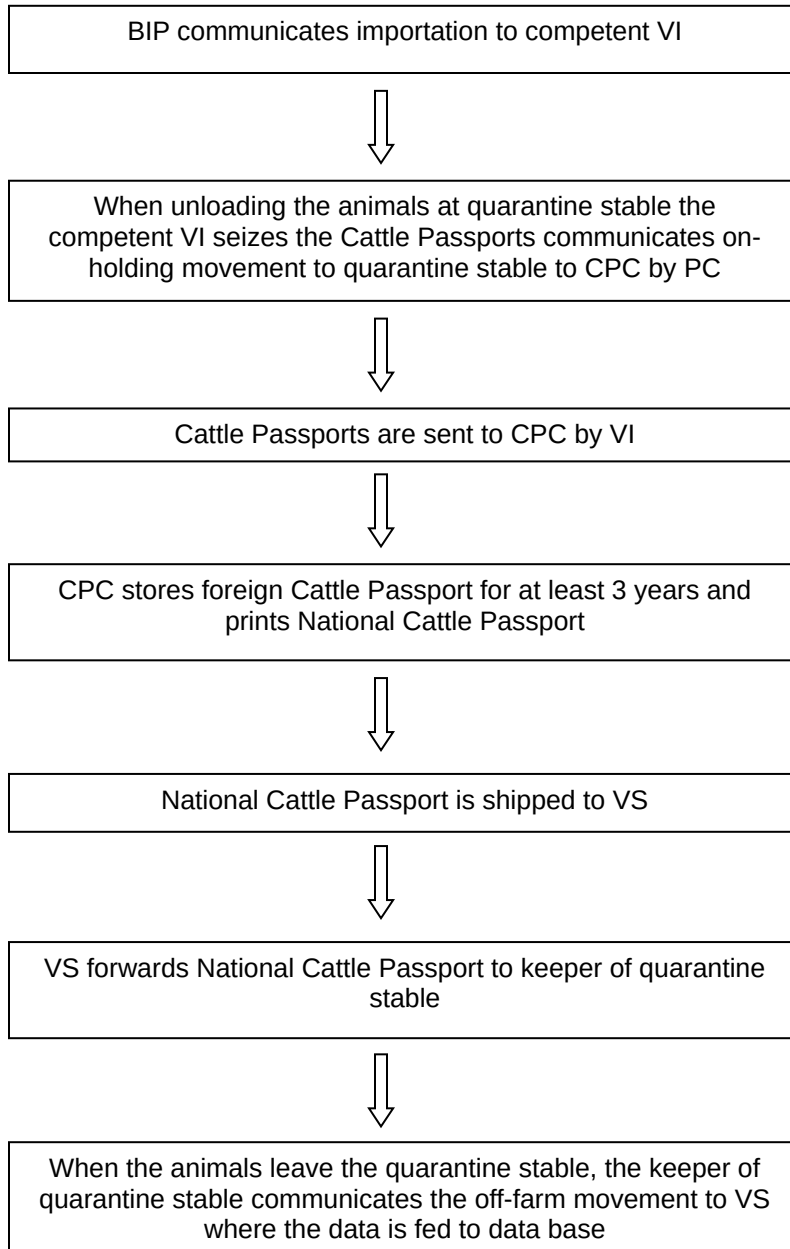
4. Print an on-the-spot control form

- Contains a list of animals currently on holding
- Basic information on each animal is included: birth date, sex, breed, date of on-movement
- Added are check boxes: fully tagged, partly tagged, untagged; no passport; not included in farm register, not present on holding
- Added at the end are empty lines to include animals found on holding but not registered in central register.

5. Enter results of on-the-spot control

Instance no. 19: Importation of animals from countries following EU rules

Working scheme of importations of animals from countries following EU rules



Description of importations of animals from countries following EU rules

In case of importation of animals from countries following EU rules the animal ID no remains unchanged. When the Importer passes the border the BIP communicates importation to competent VI.

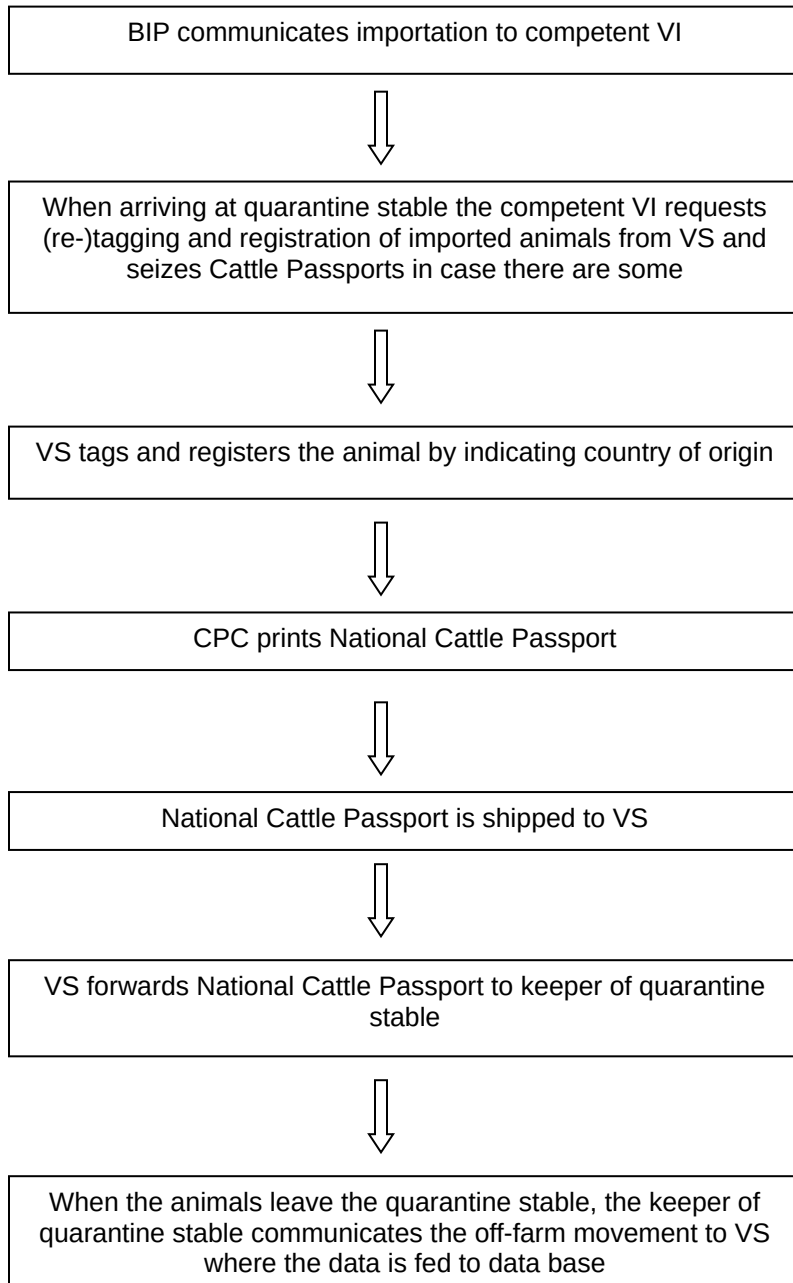
When the animal arrive at quarantine stable the VI seizes cattle passports and communicates on-holding movement to quarantine stable by indicating country of origin. A new national Cattle Passport is printed by CPC by indicating country of origin. The new Cattle Passport is sent to VS.

Establishment of a National Animal Identification System MA 9904-01-02

When the animals leave the quarantine stable, the keeper of the quarantine stable communicates the off-farm movement to VS, where the data is fed to the database.

Instance no. 20: Importation of animals not intended for slaughter from 3rd countries

Working scheme of importations of animals not intended for slaughter from 3rd countries



Description of importations of animals from countries following EU rules

When the Importer passes the border the BIP communicates importation to competent VI.

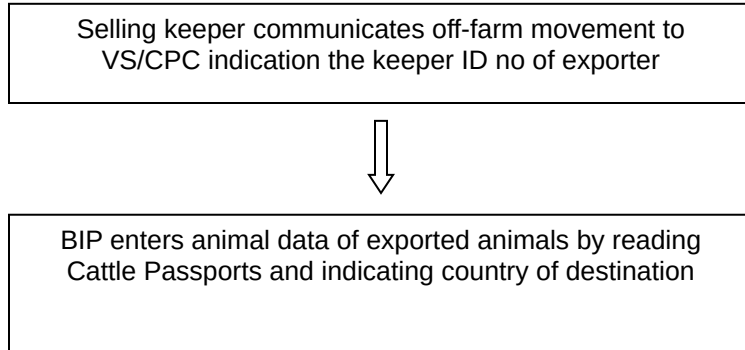
Establishment of a National Animal Identification System MA 9904-01-02

When the animal arrive at quarantine stable the VI requests registering and tagging of animals from VS and seizes cattle passports in case there are some.

VS tags and registers animal by indicating country of origin. A national Cattle Passport is printed by CPC by indicating country of origin. The new Cattle Passport is sent to VS of the keeper. When the animals leave the quarantine stable, the keeper of the quarantine stable communicates the off-farm movement to VS, where the data is fed to the database.

Instance no. 21: Exportation of animals

Working scheme of exportations of animals



Description of exportations of animals

In case of exportation the most recent keeper communicates off-farm movement to CPC through VS. When the animal passes the border, the BIP reads the cattle passport and enters an off-country movement to central database indicating the most recent keeper and country of origin when the animal passes the border.

Instance no. 22: Error correction procedure: spotted in field

An error is spotted in the field (by keepers or vet)

1. Discrepancies btw central database (+passport) and actual situation on a particular animal
 - o identification number in the tag does not match the one in the passport
 - o sex/breed/dam is different

Action: the vet collects the passport, marks correct data in the passport, signs the corrections and sends the passport to CPC (if a passport is not available/lost, the keeper produces a written declaration stating so; the vet marks corrections on the declaration)

CPC runs plausibility checks on new data

If ok, corrections are made, and a replacement passport is printed

If not ok, problems are communicated to VS asking to clear the situation.

2. Movement errors: an on/off movement has not been reported (=same as Instance 11)

a) death, slaughter in slaughterhouse, home slaughter, buy, sell

Action: the VS attempts to resolve the situation by entering an appropriate movement

If not successful, they will communicate the problems to the CPC via a written note

b) buy via market, sell via market

Instance no. 23: Error correction procedure: spotted centrally at data entry (a priory plausibility checks)

After the data from the PDA are uploaded and before they are inserted into the central database they are subjected to plausibility checks. Records that pass the checks are inserted into the database. Rejected records are equipped with reasons for rejection.

The technicians in CPC review the rejected records:

- A. Cases that can be resolved by the technician are resolved and entered into the database.
- B. Cases that require further clarification are investigated. Other relevant parties are contacted for more information. After being resolved, and the record is keyed into the database. The record is equipped with the code of technician, timestamp, and the case archive number (if applicable). Any documentation produced during the process is archived (written documents, e-mail, remarks on telephone conversation).
- C. Complex cases which can not be resolved according to regular procedures are handed to the responsible person within CPC who decides in the case. If necessary, the case is handed to the VI for on-spot control.

Instance no. 24: Error correction procedure: spotted centrally at regular intervals (posteriori plausibility checks)

Regular data consistency checks are run in the central database. Among the main checks are ones checking for completeness of movement reporting:

- Does an on-movement have a corresponding off-movement recorded
- Does an off-movement have a corresponding on-movement recorded
- Do dates match
- Do holding numbers match
- Are animals older than 10 years?

An error list is produced and assigned to the relevant VS; the list is sorted by villages or dates within the VS area.

The VS attempts to resolve problems during visits on farm. Unresolved problems are communicated back to CPC.

Instance no. 25: The I&R Archive

The archive consists of the central I&R archive and the local archives at the VS.

The central archive consists of:

- the passports of ceased animals
- all correspondence
 - o on eartag allocation
 - o on cases assigned to VI (routine control)
 - o on special cases assigned to VI
 - o on measures taken by the VI (with regards to I&R)
- any other relevant documentation

Each archive entry is assigned a unique archive number which is linked to the physical placing of the document in the storage place.

Each of the VS is obliged to maintain the I&R archive according to common specifications. The VS archives consist of:

- the holding/keeper census forms
- the change of holding/keeper forms
- the animal census (first tagging) forms
- the animal tagging receipts
- the order forms

Each of the VI is obliged to maintain the I&R archive according to common specifications.

The VI archives consist of:

- o on cases assigned to the VI as routine control according to risk analysis
- o on special cases assigned to the VI by CPC
- o on any other cases with regard to I&R
- o on measures taken by the VI (with regard to I&R)

Each of the BIP is obliged to maintain the I&R archive according to common specifications.